Customer Shopping Behavior Analysis

1.Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
– Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
 - Missing Data: 37 values in Review Rating column

3.Exploratory Data Analysis (EDA)

1. Data Loading & Initial Exploration

- Imported dataset using **pandas**
- Used:
  - df.info() → checked data types & null values
  - df.describe() → summary statistics
- Verified dataset structure (3,900 rows, 18 columns)

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2. Data Cleaning

- Identified missing values in **Review Rating**
- Handled missing data using:
  - **Median imputation (by product category)**
- Ensured data consistency across all columns

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3. Data Transformation

- Renamed columns to **snake_case format** for readability
- Standardized column names for easier SQL integration

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4. Feature Engineering

- Created new features to enhance analysis:
  - **Age Group** → categorized customers:
    - Young Adult
    - Adult
    - Middle-aged
    - Senior
  - **Purchase Frequency (Days)** → derived from purchase behavior

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5. Data Validation

- Checked for redundant columns:
  - Found **promo_code_used** redundant with discount column
  - Removed unnecessary columns
- Ensured no duplicates or inconsistent entries

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6. Key Observations from EDA

- Customer data is well distributed across categories
- Majority of customers fall into **Adult & Middle-aged groups**
- Some missing ratings required careful handling
- Discounts and purchase behavior show strong correlation

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7. Data Preparation for SQL

- Cleaned dataset exported and loaded into **PostgreSQL**
- Ensured:
  - Proper data types
  - No null inconsistencies
  - Ready for advanced SQL analysis

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	99

Total rows: 839 Query complete 00:00:

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

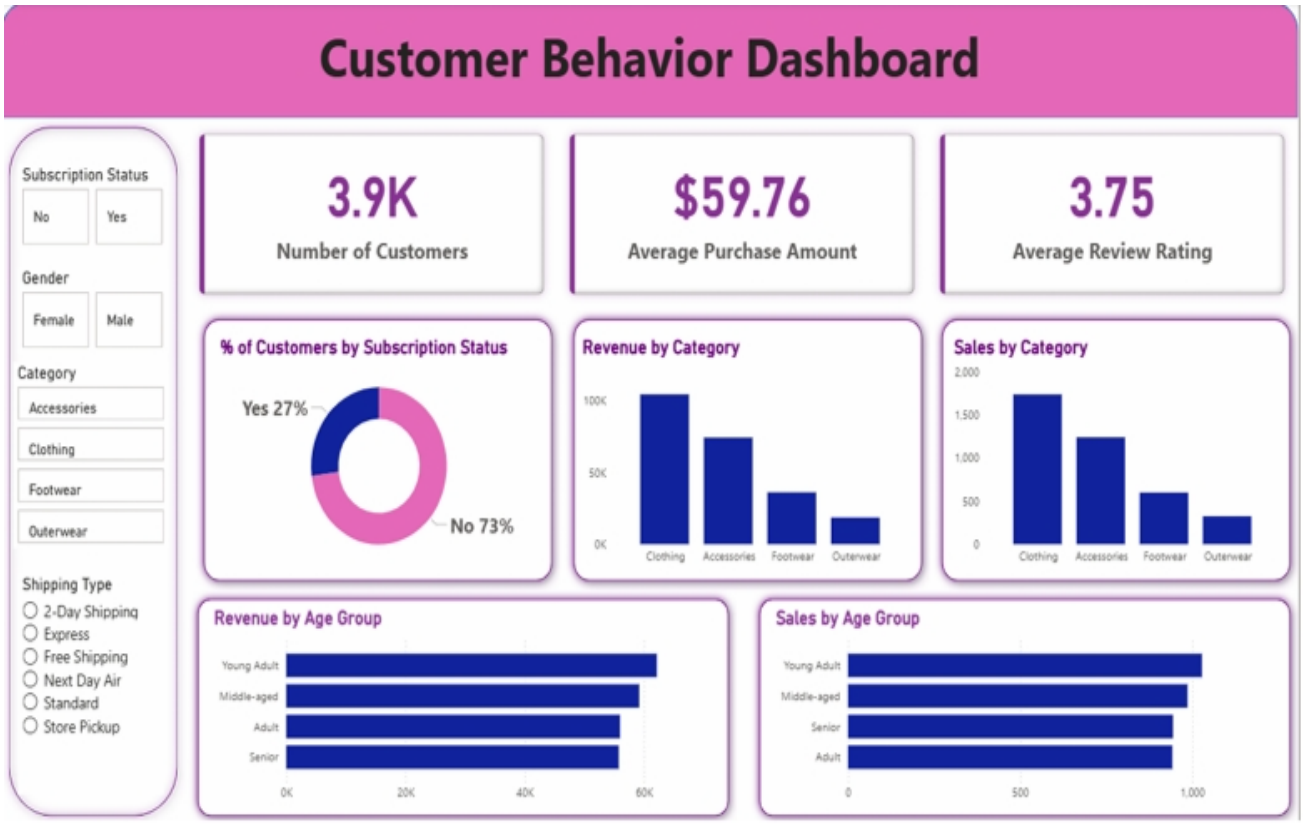
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

● **Boost Subscriptions** – Promote exclusive benefits for subscribers.

● **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.

● **Review Discount Policy** – Balance sales boosts with margin control.

● **Product Positioning** – Highlight top-rated and best-selling products in campaigns.

● **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.